

Computer Science & IT

Theory of Computation



Context Free Languages

Lecture No. 3



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Recap of Previous Lecture



Topic

CFG

Topic

Topic

Topics to be Covered



Topic

Topic

Topic

CFG

$$L = \{a^n b^n / n \geq 0\} \neq a^* b^*$$

CFG:

$$S \rightarrow a S b / \epsilon$$

S
|
ε



$a^n b^n$

$$L = \{ \underline{a^m b^m} \underline{c^n d^n} / m, n \geq 0 \}$$

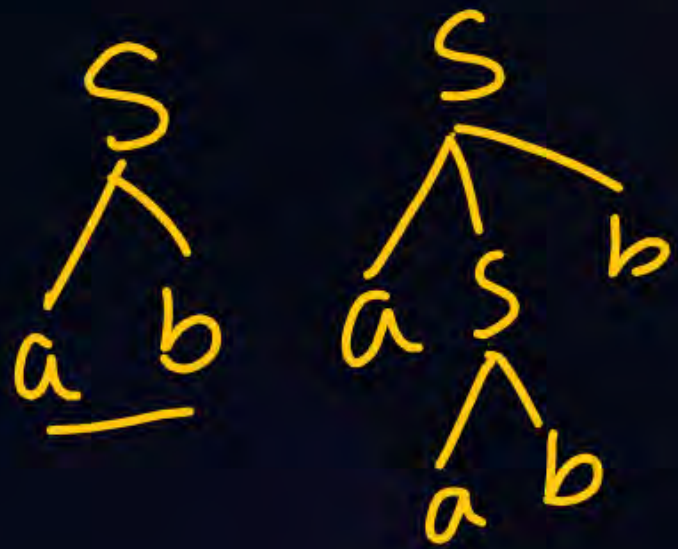
$$S \rightarrow AB$$

$$A \rightarrow aAb / \epsilon \Rightarrow a^m b^m$$

$$B \rightarrow cBd / \epsilon \Rightarrow c^n d^n$$

$$a^n b^n / n \geq 1$$

$$S \rightarrow \underline{a} \underline{S} \underline{b} / \underline{a} \underline{b} \checkmark$$



$$a^n b^n / n \geq 2$$

$$S \rightarrow a \overset{\checkmark}{S} b / a \overset{\checkmark}{\underline{a}} b b$$

3 min time

$$s \rightarrow a s b / \underline{a b b}$$

$$s \rightarrow \underline{a s b} \Rightarrow \underline{a}^n \underline{s} \underline{b}^n$$

$$\Rightarrow a^n \downarrow a b b b^n$$

$$\Rightarrow \underline{\underline{a^{n+1} b^{n+2}}} / \underline{\underline{n \geq 0}}$$

$$S \rightarrow \underline{aSbb} / \underline{\underline{aab}}$$

$$S \rightarrow aSbb \Rightarrow \underline{a^n} S \underline{b^{2n}}$$

$$a^n a a b b^{2n}$$

$$a^{n+2} b^{2n+1} / n \geq 0 \checkmark$$

$$a^{2n}b^n / n \geq 0$$

$$\underline{S \rightarrow \underline{a}aSb / \epsilon}$$

$$\left\{ \begin{array}{l} a^n b^{2n} \rightarrow \text{same} \\ a^{10n} b^n \rightarrow \text{in} \\ a^n b^{100n} \rightarrow \end{array} \right.$$

3 min

$$a^n b^{2n+3} / n \geq 0$$

$$a^n b^3 b^{2n}$$

$$\underline{a}^n \underline{bbb} \underline{b}^{2n}$$

$$S \rightarrow a \underline{S} b b / b b b$$

$$a^n b^{3n+4} / n \geq 0$$

$$\underline{a^n} \underline{b b b b} \underline{b^{3n}}$$

$$S \rightarrow a S b b b / b b b b$$

$$a^{3n+2} b^n / n \geq 0$$

$$a^{3n} a^2 b^n$$

$$a^{3n} \underline{aa} b^n$$

$$S \rightarrow \underline{aaa} S \underline{b} / aa$$

$$a^{2n}b^{3n}/n \geq 0$$

$$S \rightarrow aaSbbb/\epsilon$$

$$\underline{a}^{2n+3} \underline{b}^{3n+2}/n \geq 0$$

$$a^{2n}a^3b^2b^{3n}$$

$$\underline{a}^{2n}aaa \underline{b}b \underline{b}^{3n}$$

$$S \rightarrow \underline{a}a \underline{bbb}/aaa \underline{bb}$$

$$\checkmark \underline{a^m b^n} / \underline{m \geq n} / \underline{m, n \geq 0} \quad 3 \text{ min}$$

$$\overset{0}{a^*} a^n b^n \checkmark$$

$$\begin{aligned} S &\rightarrow BA \\ A &\rightarrow aAb / \epsilon \\ B &\rightarrow aB / \epsilon \end{aligned}$$

$$\Rightarrow a^n b^n$$

$$\Rightarrow a^*$$

$$\equiv \checkmark$$

$$\underline{a^m b^n} / m > n / m, n \geq 0$$

$$a^+ \underline{a^n b^n}$$

$$S \rightarrow BA$$

$$A \rightarrow aAb / \epsilon \Rightarrow a^n b^n$$

$$B \rightarrow aB / a \Rightarrow a^+$$

$$a^m b^n / m \leq n$$

$$a^n b^n b^*$$

$$S \rightarrow AB$$

$$\checkmark A \rightarrow aAb / \epsilon \Rightarrow a^n b^n$$

$$B \rightarrow bB / \epsilon \Rightarrow b^*$$

$$a^m b^n / m < n$$

$$\underline{a^n} \underline{b^n} b^+$$

$$S \rightarrow AB$$

$$A \rightarrow aAb / \epsilon \Rightarrow a^n b^n$$

$$B \rightarrow bB / b \Rightarrow b^+$$

3min

$$a^m b^n / m \neq n$$

$$\underline{a^+} \underline{b^n} \cup \underline{a^n} \underline{b^+}$$

$$S \rightarrow \underline{B} \underline{A} / AC$$

$$A \rightarrow aAb / \epsilon \Rightarrow a^n b^n \checkmark$$

$$B \rightarrow aA / a \Rightarrow a^+ \checkmark$$

$$C \rightarrow bC / b \Rightarrow b^+ \checkmark$$

3 min



$$S \rightarrow \underline{AS}_1 / S_1 B$$

$$S_1 \rightarrow aS_1 b / \epsilon \Rightarrow a^n b^n$$

$$A \rightarrow aA / \epsilon \Rightarrow a^*$$

$$B \rightarrow bB / b \Rightarrow b^+$$

$$a^* a^n b^n + a^n b^n b^+ \\ a^n b^m \cup a^n b^m \\ \underline{\underline{n \geq m}} \quad \underline{\underline{n < m}}$$

$$\underline{a^* b^+}$$

$$a^m b^n / m \geq 2n$$

$$\underline{a^* a^{2n} b^n}$$

$$S \rightarrow BA$$

$$A \rightarrow a^n b$$

$$B \rightarrow aB / \epsilon$$

$$a^m b^n / m \geq 2n+3$$

$$\underline{\underline{a^* a^{2n+3} b^n}}$$

$$S \rightarrow BA$$

$$A \rightarrow aaAb / aaa \Rightarrow a^{2n+3} b^n$$

$$B \rightarrow aB / \epsilon \checkmark$$

$$a^m b^n / m \leq 2n + 3$$

↓
It is difficult to write CFG for this language

But this is a CFL

$$\underline{\underline{a^n b^n}} \cup \underline{\underline{a^n b^{2n}}}$$

$$S \rightarrow A/B$$

$$A \rightarrow aAb/\epsilon \Rightarrow a^n b^n$$

$$B \rightarrow aBbb/\epsilon \Rightarrow a^n b^{2n}$$

$$a^m b^n / m \leq n \leq 2m \rightarrow \text{CFL}$$

$$[S \rightarrow asb / asbb / \epsilon]$$

Difficult to explain ✓

$$\Sigma = \{a, b, c\}$$

$$\underline{a^m b^m} \subseteq^n / m, n \geq 0$$

$$S \rightarrow AB$$

$$A \rightarrow aAb / \epsilon$$

$$B \rightarrow cB / \epsilon$$

$$\underline{\underline{a^n b^n c^*}}$$

$$= \underline{\underline{a^m b^n c^n}} / m, n \geq 0$$

$$a^* b^n c^n$$

$$S \rightarrow BA$$

$$A \rightarrow bAc / \epsilon$$

$$B \rightarrow aB / \epsilon$$

$$a^n b^m c^n$$

$$\underline{a}^n b^* \underline{c}^n$$

$$S \rightarrow \underline{a} S \underline{c} / \underline{B}$$

$$B \rightarrow bB / \epsilon$$

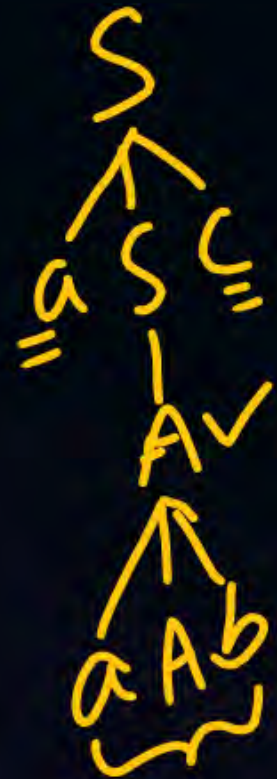
$$a^m b^n c^p / m = n + p$$

$$a^{n+p} b^n c^p$$

$$a^p \underline{a^n} b^n c^p$$

$$S \rightarrow \underline{a} S c / \underline{A} \checkmark$$

$$A \rightarrow \underline{a} A b / \epsilon$$



$$a^m b^n c^p / p = m + n$$

$$a^m b^n c^{m+n}$$

$$\underbrace{a^m b^n c^n}_{c^m} c^m$$

$$S \rightarrow aSc / A$$

$$A \rightarrow bAc / \epsilon$$

$$a^m b^n c^p / n = m + p$$

$$a^m b^{m+p} c^p$$

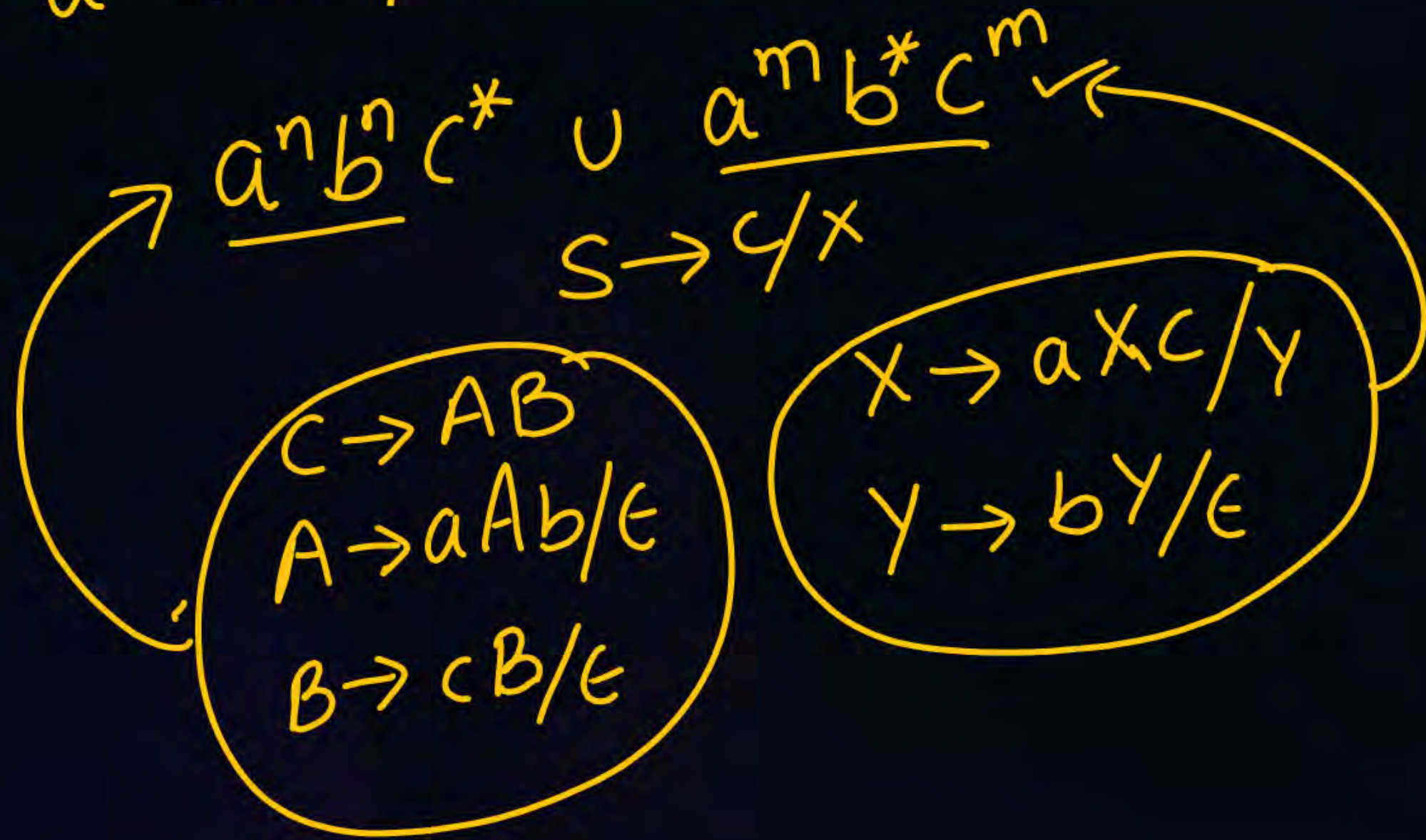
$$\underline{a^m b^m} \underline{b^p c^p}$$

$$S \rightarrow AB$$

$$A \rightarrow aAb / \epsilon$$

$$B \rightarrow bBc / \epsilon$$

$$a^m b^n c^p / m=n \vee m=p$$



$$\underline{a^m b^n c^p d^q} / m=n \text{ and } p=q$$

$$S \rightarrow AB$$

$$A \rightarrow aAb / \epsilon$$

$$B \rightarrow cBd / \epsilon$$



3 questions $\rightarrow$ 5 min



Note:- alternate Comparison is not CFL
alternate addition is CFL

$a^m b^n c^p d^q$

① $m+n=p+q$

② $m+q=n+p$

③ $m+p=n+q$

CFL

difficult to write CFG

$\times a^m b^n c^m d^n \rightarrow \text{XCFL} \quad \checkmark \text{CSL}$

$\checkmark a^m b^n c^p d^q (m+p=n=q) \rightarrow \text{CFL}$

$$L = \{w / n_a(w) = n_b(w)\} \neq a^n b^n \checkmark$$

$\Downarrow$
 $aabb \quad \underline{abab} \quad \underline{bbaa}$

$$S \rightarrow \underline{a}S\underline{b} / \underline{b}S\underline{a} / SS / \epsilon$$

$$S \rightarrow aS\underline{bS} / \underline{bSaS} / \epsilon$$

$$S \rightarrow SaSb / SbSa / \epsilon$$

$$L = \{ w / n_a(\underline{w}) = 2 * n_b(\underline{w}) \}$$

aab aba baa

$$S \rightarrow aSaSb / aSbSa / bSaSa / \underline{SS}^x / \epsilon$$

$$S \rightarrow aSaSb\underline{S} / aSbSa\underline{S} / bSaSa\underline{S} / \underline{\epsilon}$$

$$S \rightarrow \underline{S}aSaSb / \underline{S}aSbSa / \underline{S}bSaSa / \underline{\epsilon}$$

$$L = \{w / n_a(w) = 3 n_b(w)\}$$

aaab aaba abaa baac

$$S \rightarrow \underline{aSaSb / aSasbsa / asbsaSa / bsasasa / \underline{\underline{SS}} / \epsilon}$$

$$S \rightarrow aSasasbS / aSasbsaS / asbsaSaS / bsasasaS / \epsilon$$

$$S \rightarrow SaSasb / Sasasbsa / SasbsaSa / SbSasasa / \epsilon$$

$$w / n_a(w) = 100 n_b(w) \checkmark$$

min ✓

$$n(b)w = \underline{\underline{100 n_a(w)}} \checkmark$$



THANK - YOU